

Sir C. V. Raman

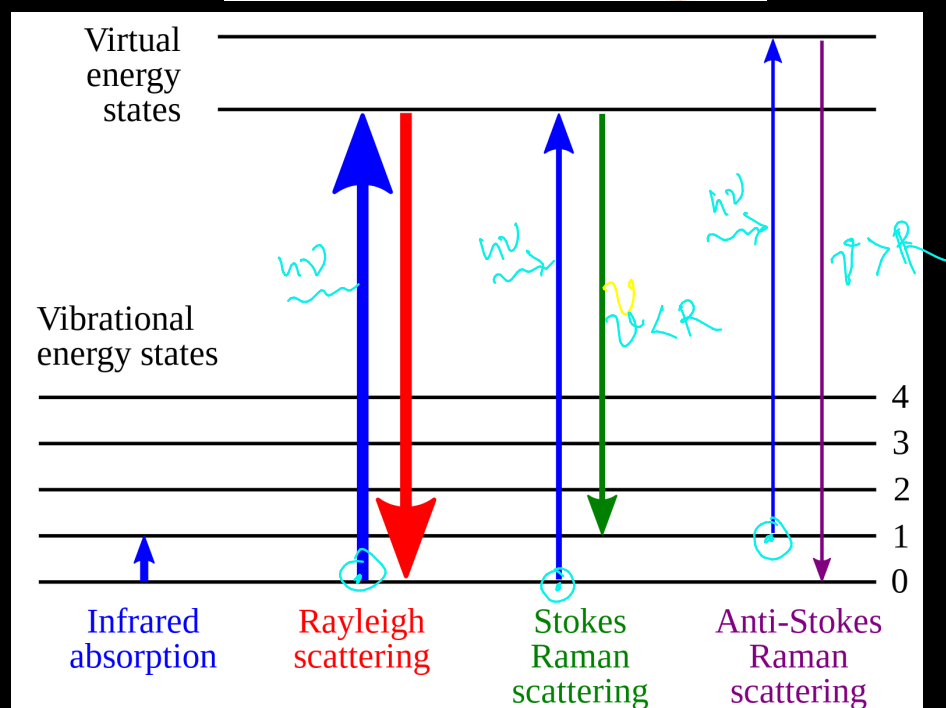
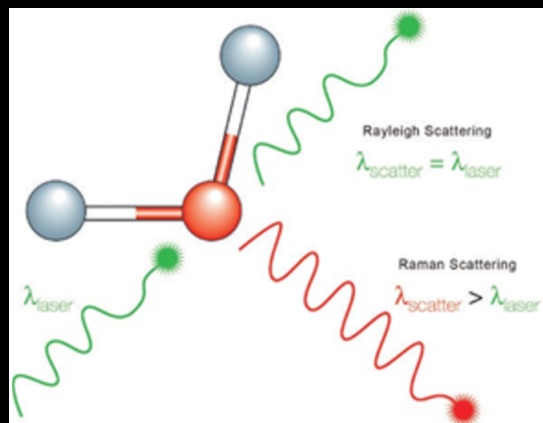
NL-1930 - only one from India in Science, in physics.



- Originally sunlight was used
- LASER is used now.

Experiment was done at Indian Association for the Cultivation of Science (IACS), Kolkata

Principle of Raman Spectroscopy:



Rayleigh Scattering

$E_{\text{Antistokes}} > E_{\text{Rayleigh}} > E_{\text{Stokes}}$

Stokes Lines

Anti-stokes Lines

$\Delta E \leftarrow$ Stokes Lines

Anti-stokes Lines $\rightarrow \Delta E$

ΔE is equal

$\Downarrow$

Raman Shift

unit $\Rightarrow \text{cm}^{-1}$

$\lambda_{\text{antistokes}} < \lambda_{\text{Rayleigh}} < \lambda_{\text{stokes}}$

Q: $\lambda_{ex} = 5650 \text{ \AA}$ and Stokes line is $5810 \text{ \AA}$. Calculate the wavelength of anti-Stokes line.

Hints: $E = h\nu = \frac{hc}{\lambda} = hc\bar{\nu}$

Vibrational Raman Spectroscopy

$$E_v = \left(v + \frac{1}{2}\right) \bar{\omega} - \left(v + \frac{1}{2}\right)^2 \bar{\omega} x_e$$

Selection Rule, $\Delta v = 0, \pm 1, \pm 2, \dots$

$$\Delta E_{v=0 \rightarrow 1} = \omega(1 - 2x_e)$$

$$\bar{\nu}_{\text{antistokes final}} = \bar{\nu}_{\text{initial}} + \bar{\omega}_e(1 - 2x_e)$$

$$\bar{\nu}_{\text{stokes final}} = \bar{\nu}_0 - \bar{\omega}_e(1 - 2x_e)$$

$$\boxed{\bar{\nu} = \bar{\nu}_0 \pm \bar{\omega}_e(1 - 2x_e)}$$

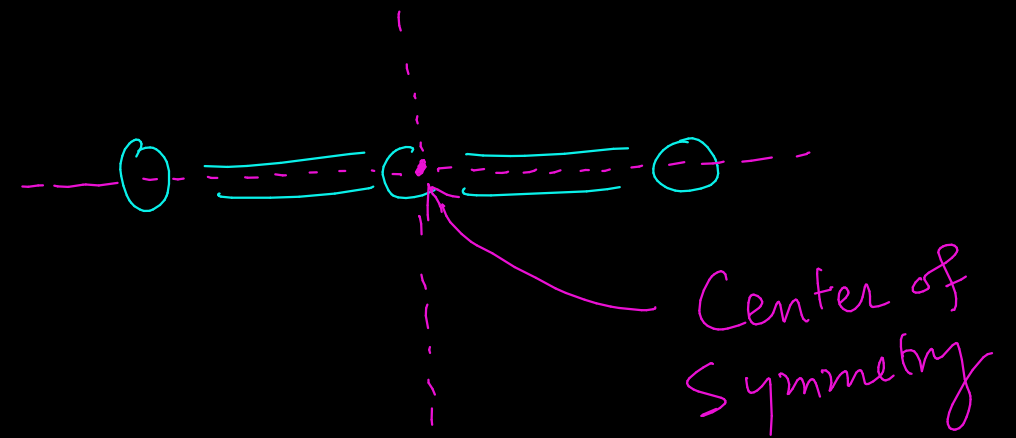
+ $\Rightarrow$ A. Stokes
- $\Rightarrow$ Stokes

Rules of Mutual Exclusion Principle

Table 4.1 Raman and infra-red activities of carbon dioxide

Mode of vibration of CO ₂	Raman	Infra-red
ν_1 : symmetric stretch	Active	Inactive
ν_2 : bend	Inactive	Active
ν_3 : asymmetric stretch	Inactive	Active

Rule of mutual exclusion. If a molecule has a *centre of symmetry* then Raman active vibrations are infra-red inactive, and vice versa. If there is no centre of symmetry then some (but not necessarily all) vibrations may be both Raman and infra-red active.



- If there is no center of symmetry then, IR/Raman will not alternate.

